CAS SciFinder

Task History

Initiating Search

November 5, 2025, 8:22 PM

Search:

CAS Lexicon Search:

Steglich esterification - Preferred Concept

Search Tasks

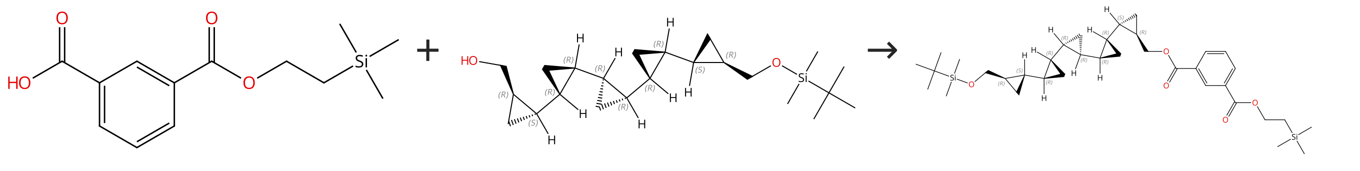
Task	Result Type	View
Returned Reference Results from CAS Lexicon (360)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (891) Filtered By: Document Type: **Journal** Language: **English**	**Reactions**	View Results

CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)

31-356-CAS-4461896 Steps: **1** Yield: **100%**

1.1 **Reagents:** *sec*-Butyllithium, 4-(Dimethylamino)pyridine
Solvents: Dichloromethane

1.2 **Solvents:** Dichloromethane, Water

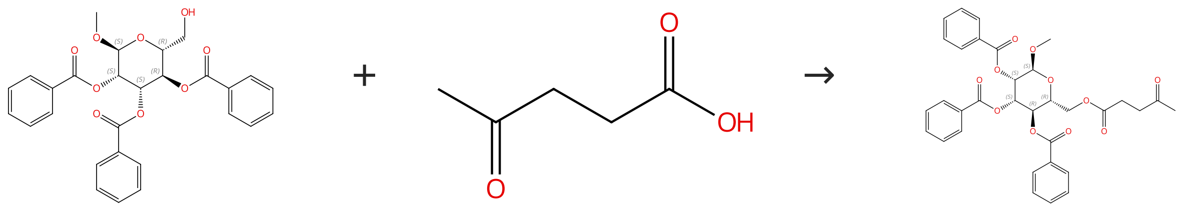
Experimental Protocols

Synthesis and Characterization of Coronanes: Multicyclopropane-Fused Macrocyclic Arrays

By: Barrett, Anthony G. M.; et al

Journal of Organic Chemistry (2001), 66(7), 2187-2196.

Scheme 2 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (98)

Absolute stereochemistry shown

31-614-CAS-35529987 Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine; 30 min, 100 °C

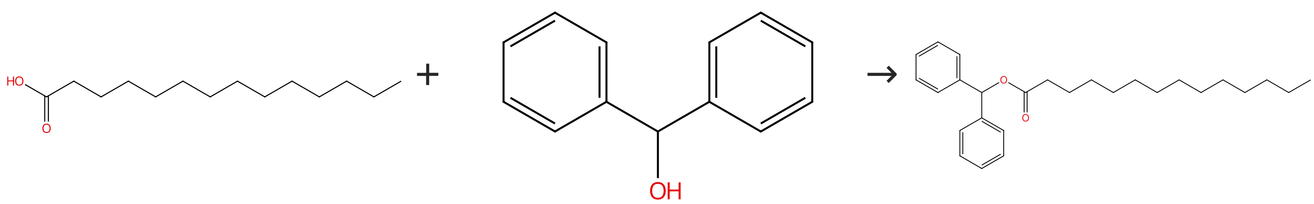
Experimental Protocols

A comprehensive solvent-free approach for the esterification and amidation of carboxylic acids mediated by carbodiimides

By: Traboni, Serena; et al

Tetrahedron (2023), 133, 133291.

Scheme 3 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (122)

Suppliers (97)

31-614-CAS-35529994 Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine; 30 min, 100 °C

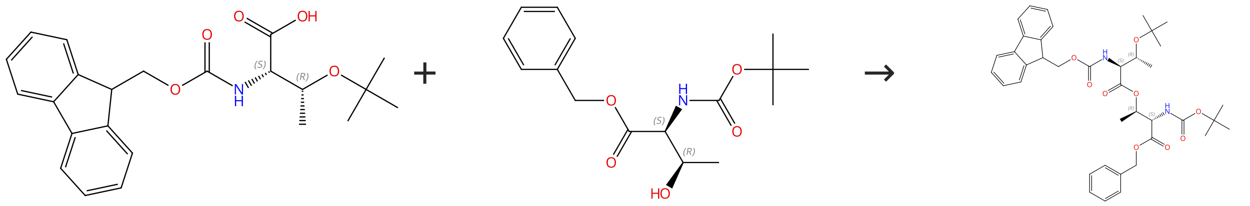
Experimental Protocols

A comprehensive solvent-free approach for the esterification and amidation of carboxylic acids mediated by carbodiimides

By: Traboni, Serena; et al

Tetrahedron (2023), 133, 133291.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Absolute stereochemistry shown — Suppliers (90)

Absolute stereochemistry shown, Rotation (-) — Suppliers (51)

Absolute stereochemistry shown

31-356-CAS-21440281 Steps: **1** Yield: **100%**

1.1 **Reagents:** Diisopropylcarbodiimide
Solvents: Dichloromethane; 5 min, 0 °C

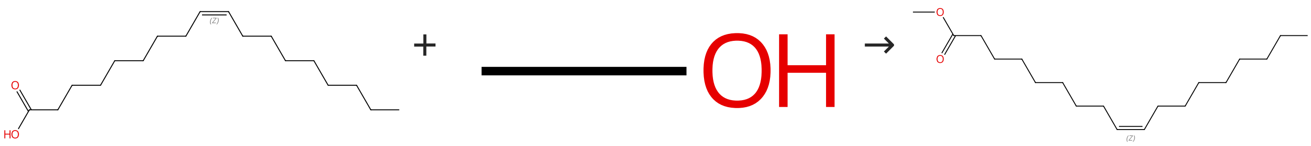
1.2 **Reagents:** 4-(Dimethylamino)pyridine
Solvents: Dimethylformamide; 2 h, 0 °C; 0 °C → rt; 18 h, rt

General Fmoc-based solid-phase synthesis of complex depsipeptides circumventing problematic Fmoc removal

By: Lobo-Ruiz, Ariadna; et al

European Journal of Organic Chemistry (2020), 2020(2), 183-192.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Double bond geometry shown — Suppliers (173)

Suppliers (444)

Double bond geometry shown — Suppliers (122)

31-356-CAS-19862902 Steps: **1** Yield: **100%**

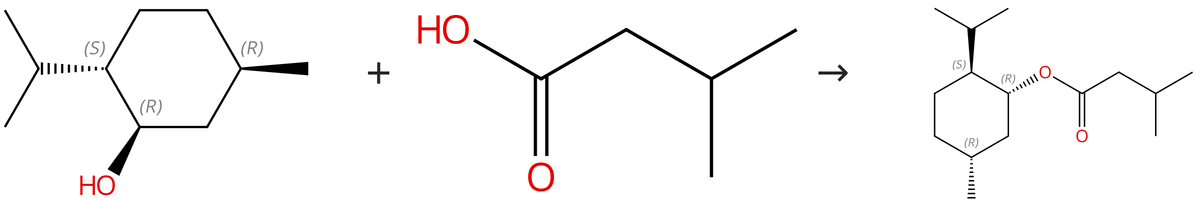
1.1 **Catalysts:** Sulfuric acid; 3 h, 90 °C

Fatty acid-based crosslinkable polymethacrylate coatings

By: Decostanzi, Melanie; et al

Progress in Organic Coatings (2018), 124, 147-157.

Scheme 6 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (-) — Suppliers (132)

Suppliers (104)

Relative stereochemistry shown — Suppliers (11)

31-356-CAS-22557470 Steps: **1** Yield: **99%**

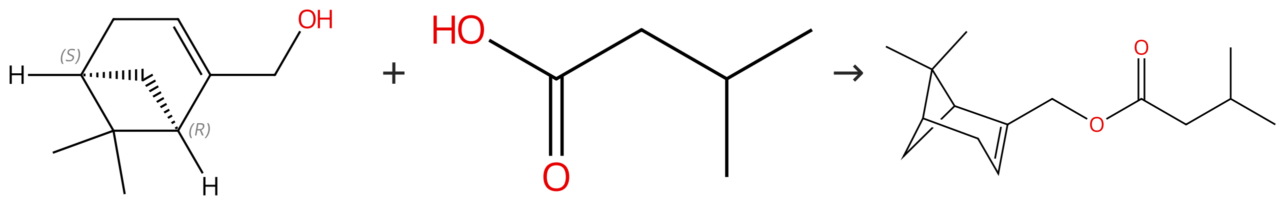
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 7 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (48) Suppliers (104) Suppliers (5)

31-356-CAS-22557471 Steps: **1** Yield: **99%**

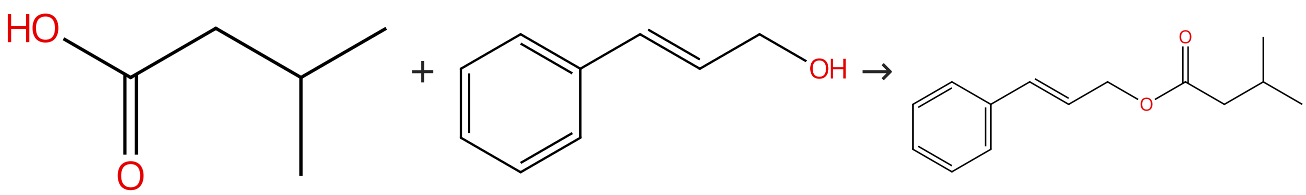
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 8 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104) Suppliers (86) Suppliers (36)

31-356-CAS-22557474 Steps: **1** Yield: **99%**

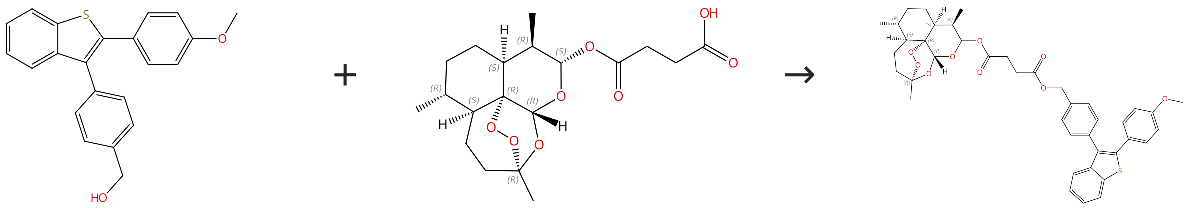
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 9 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown

Suppliers (97)

31-614-CAS-34328361 Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; overnight, 0 °C

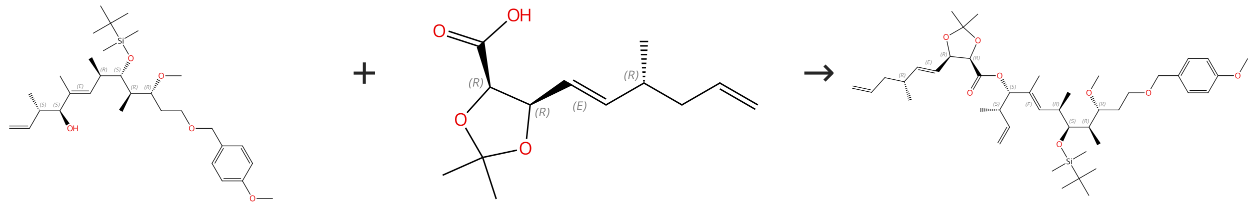
Experimental Protocols

Synthesis of novel artesunate-benzothiophene and artemisinin-benzothiophene derivatives

By: Ozok, Omruye; et al

Natural Product Research (2022), 36(20), 5228-5234.

Scheme 10 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+) Double bond geometry shown | Absolute stereochemistry shown, Rotation (-) Double bond geometry shown | Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

31-356-CAS-5762074 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; rt

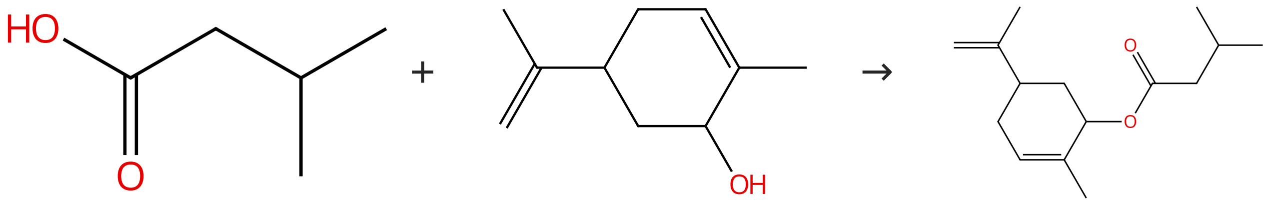
Experimental Protocols

A Concise Synthesis of Carolacton

By: Hallside, Michal S.; et al

Organic Letters (2014), 16(4), 1148-1151.

Scheme 11 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104) | Suppliers (71) | Suppliers (10)

31-356-CAS-22557479 Steps: **1** Yield: **99%**

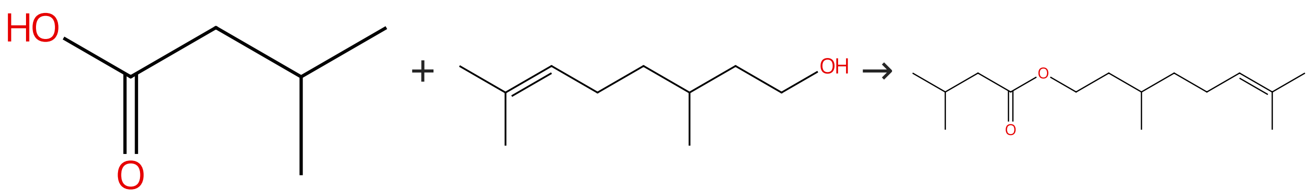
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 12 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104) | Suppliers (103) | Suppliers (9)

31-356-CAS-22557466 Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

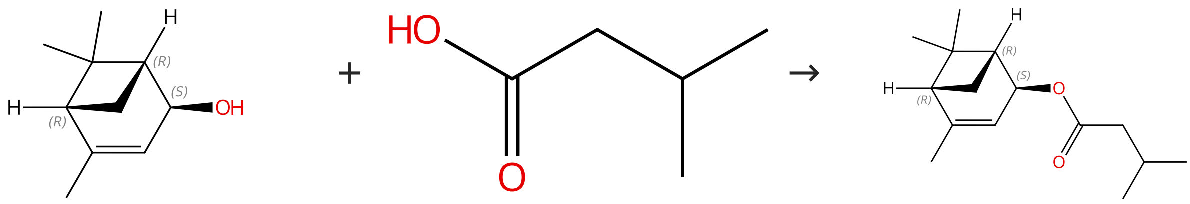
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Suppliers (12)

Suppliers (104)

Relative stereochemistry shown

31-356-CAS-22557481

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

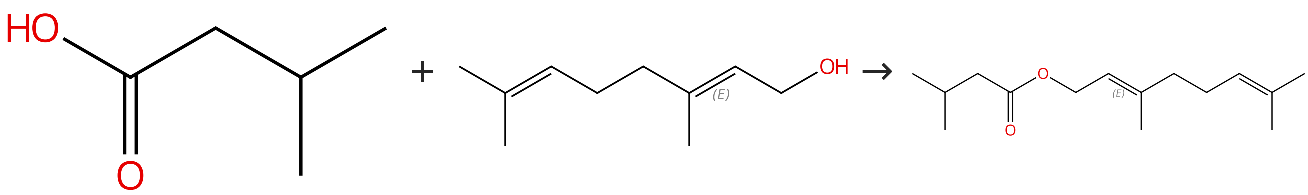
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (104)

Double bond geometry shown

Suppliers (118)

Double bond geometry shown

Suppliers (43)

31-356-CAS-22557467

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

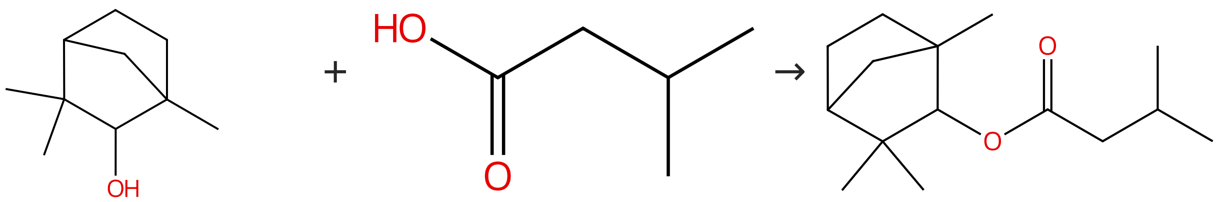
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (66)

Suppliers (104)

Suppliers (4)

31-356-CAS-22557482

Steps: **1** Yield: **99%**

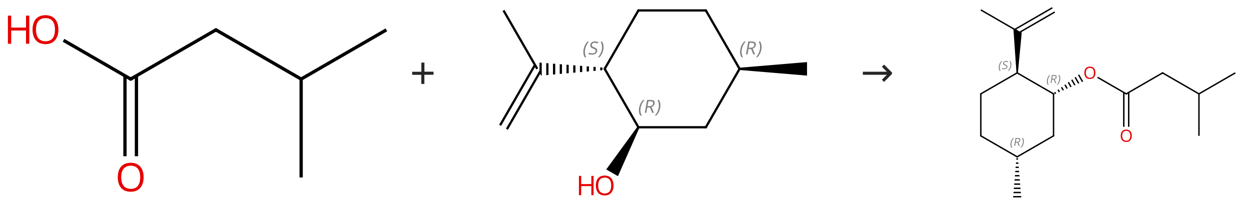
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 16 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104)

Absolute stereochemistry shown, Rotation (-)

Suppliers (65)

Absolute stereochemistry shown

31-356-CAS-22557477 Steps: **1** Yield: **99%**

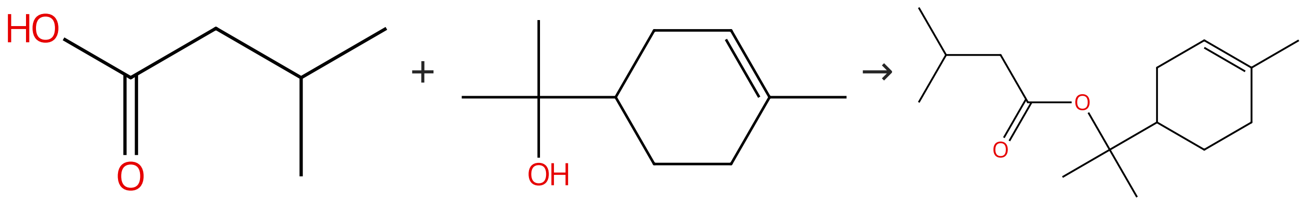
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 17 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104)

Suppliers (96)

Suppliers (9)

31-356-CAS-22557478 Steps: **1** Yield: **99%**

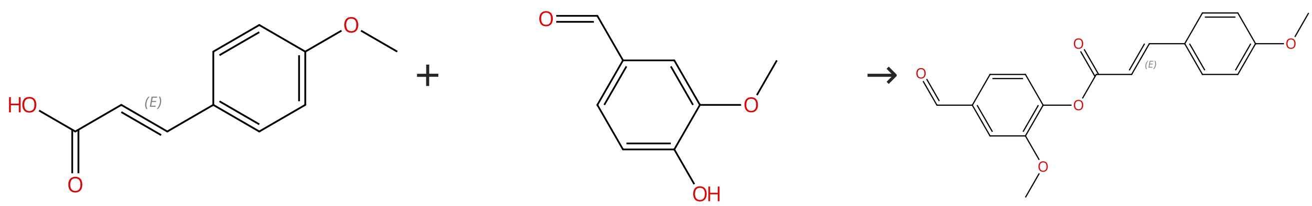
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 18 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (70)

Suppliers (146)

Double bond geometry shown

Suppliers (2)

31-614-CAS-24403408 Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 24 h, rt

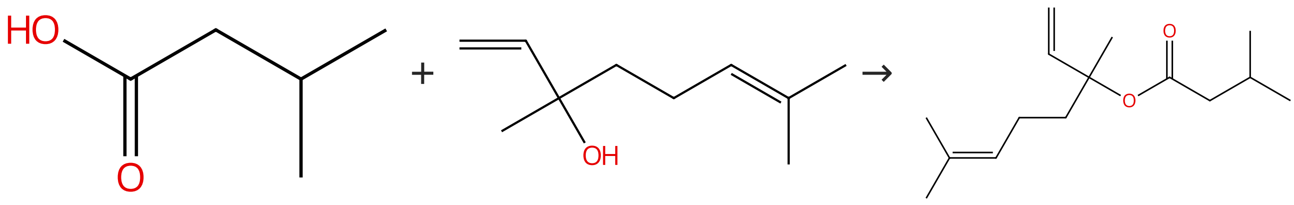
Experimental Protocols

Preparation, spectral characterization and anticancer potential of cinnamic esters

By: Goncalves, Renan O.; et al

Journal of the Brazilian Chemical Society (2021), 32(10), 1931-1942.

Scheme 19 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104) | Suppliers (111) | Suppliers (18)

31-356-CAS-22557469 Steps: **1** Yield: **99%**

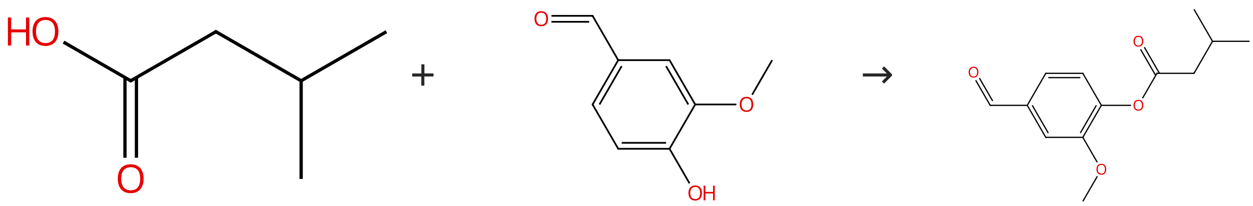
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 20 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (104) | Suppliers (146) | Suppliers (4)

31-356-CAS-22557483 Steps: **1** Yield: **99%**

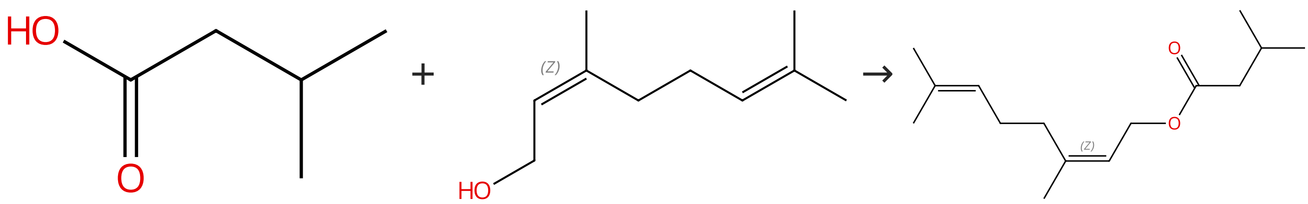
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 21 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Double bond geometry shown

Suppliers (104) | Suppliers (87) | Suppliers (23)

31-356-CAS-22557468 Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

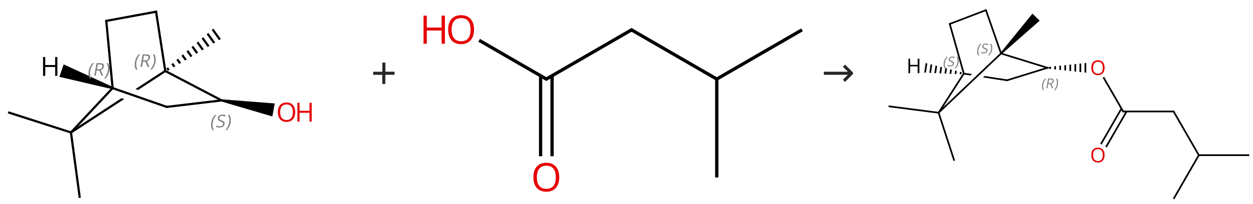
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **99%**



Relative stereochemistry shown

Suppliers (64)

Suppliers (104)

Relative stereochemistry shown

Suppliers (13)

31-356-CAS-22557472

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

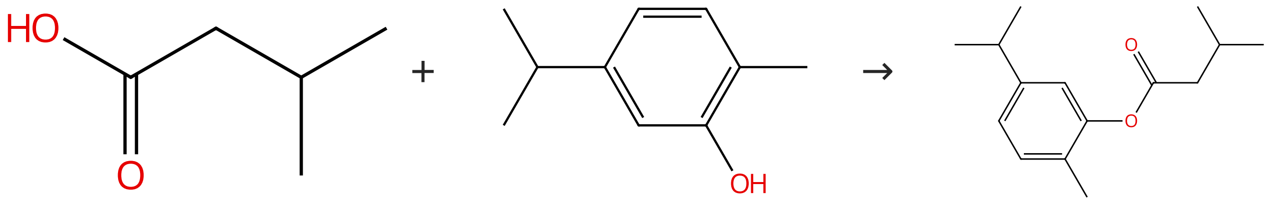
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 23 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (104)

Suppliers (103)

Supplier (1)

31-356-CAS-22557480

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

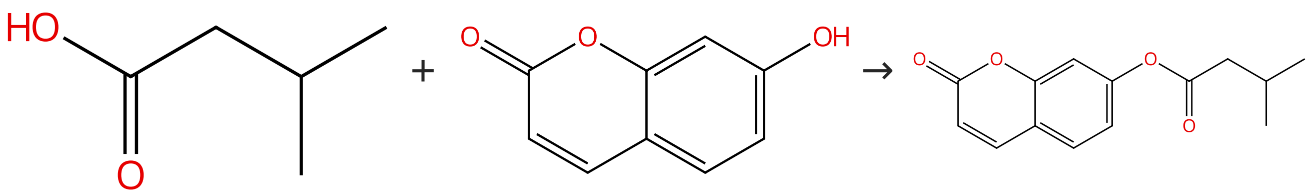
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 24 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (104)

Suppliers (113)

Suppliers (3)

31-356-CAS-22557484

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

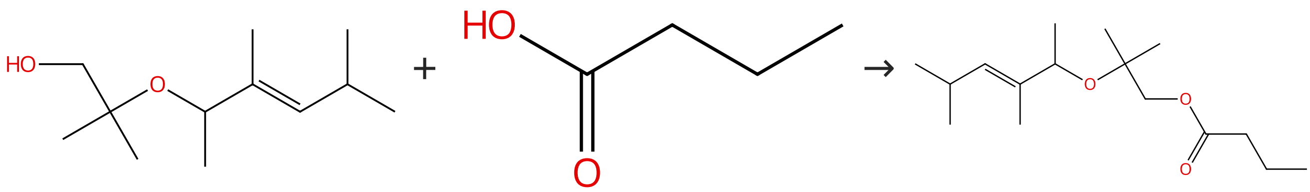
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 25 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (2) | Suppliers (82) | Suppliers (4)

31-356-CAS-14589818

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 0 °C; 1 h, rt

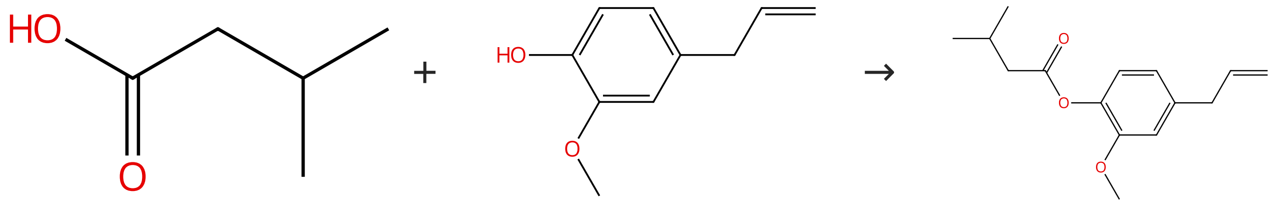
Synthesis and odor of aliphatic musks: Discovery of a new class of odorants

By: Kraft, Philip; et al

European Journal of Organic Chemistry (2004), (2), 354-365.

Scheme 26 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (104) | Suppliers (119) | Suppliers (30)

31-356-CAS-22557475

Steps: **1** Yield: **99%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

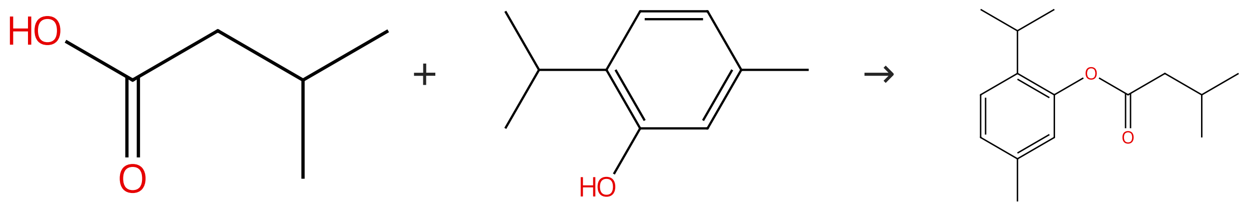
Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 27 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (104) | Suppliers (137) | Suppliers (9)

31-356-CAS-22557476

Steps: **1** Yield: **99%**

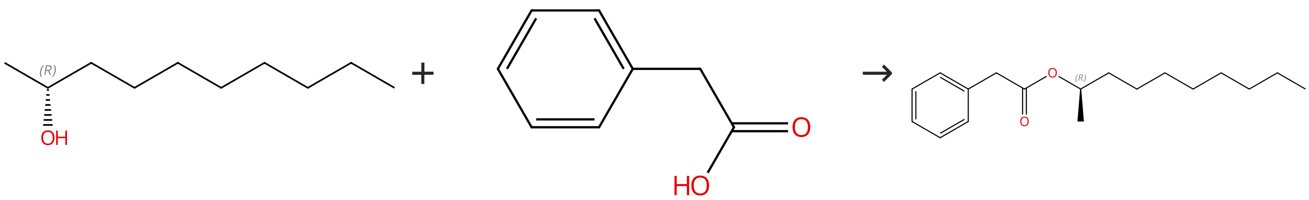
1.1 **Reagents:** Dicyclohexylcarbodiimide, Oxygen
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 1 min, 0 °C; 15 min, 0 °C; 3 h, rt

Monoterpenoid isovalerate esters as long-lasting spatial mosquito repellents

By: Klimavicz, James S.; et al

ACS Symposium Series (2018), 1289, 205-217.

Scheme 28 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (35)

Suppliers (40)

Absolute stereochemistry shown, Rotation (-)

31-356-CAS-23888663 Steps: **1** Yield: **99%**

1.1 **Reagents:** 2,6-Lutidine, 2-Chloro-1-methylpyridinium iodide
Solvents: Dimethyl carbonate; 24 h, rt

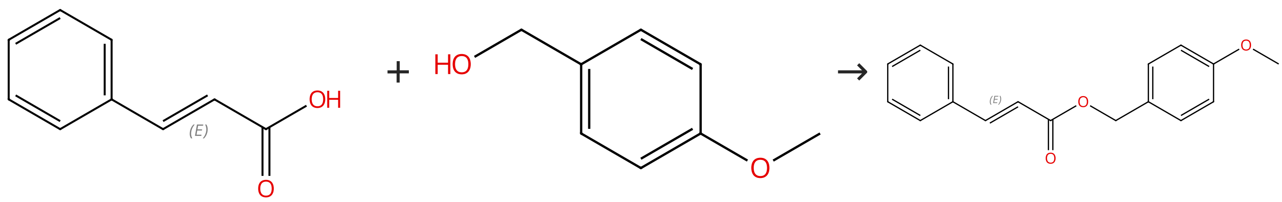
Experimental Protocols

A solvent-reagent selection guide for Steglich-type esterification of carboxylic acids

By: Jordan, Andrew; et al

Green Chemistry (2021), 23(17), 6405-6413.

Scheme 29 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Suppliers (160)

Suppliers (105)

Double bond geometry shown

Supplier (1)

31-356-CAS-19419126 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Acetonitrile; 45 min, 40 - 45 °C

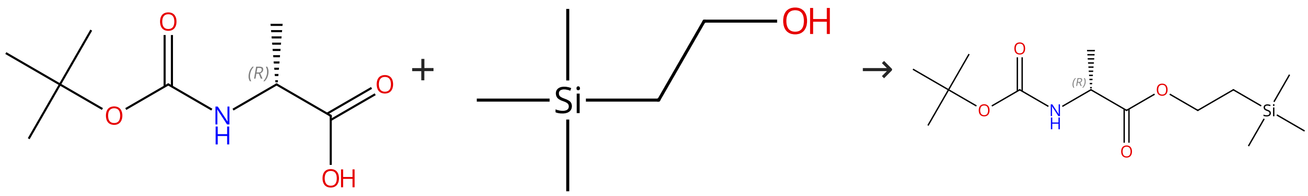
Experimental Protocols

Synthesis of (E)-cinnamyl ester derivatives via a greener Steglich esterification

By: Lutjen, Andrew B.; et al

Bioorganic & Medicinal Chemistry (2018), 26(19), 5291-5298.

Scheme 30 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Suppliers (93)

Suppliers (85)

Absolute stereochemistry shown, Rotation (+)

31-356-CAS-22375123 Steps: **1** Yield: **99%**

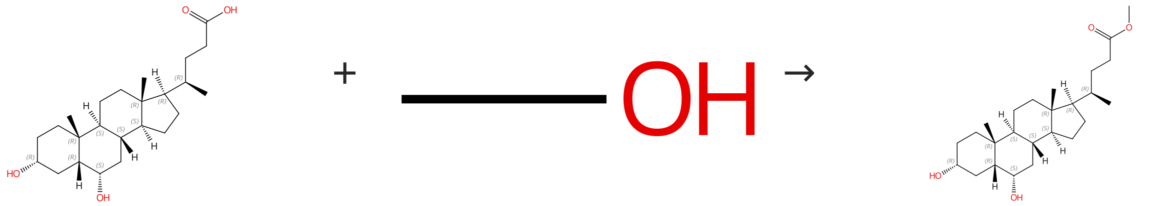
1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane, Ethyl acetate; 2 h, rt

Modular Total Synthesis of Farnesyl Analogues of Cell Wall Precursors Lipid I and II Containing the Staphylococcus aureus Pentaglycine Bridge Modification

By: Wingen, Lukas M.; et al

Journal of Organic Chemistry (2020), 85(15), 10206-10215.

Scheme 31 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown

Suppliers (76)

Suppliers (444)

Absolute stereochemistry shown, Rotation (+)

Suppliers (49)

31-614-CAS-41337688 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sulfuric acid
Solvents: Methanol; 5 min, rt; 18 h, rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; neutralized

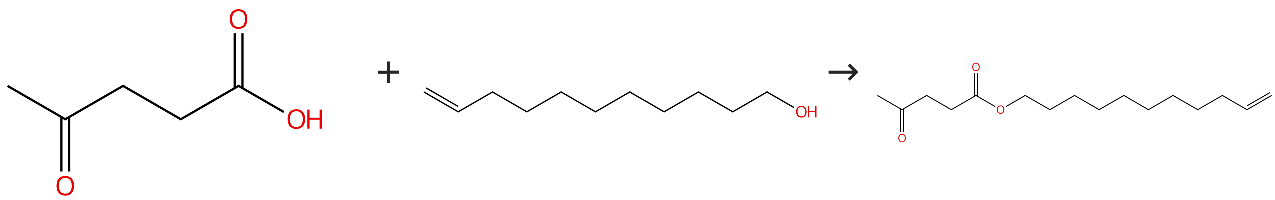
Experimental Protocols

Photoinduced decarboxylative radical cascade alkylation/cyclization of benzimidazole derivatives with an aliphatic carboxylic acid via ligand-to-iron charge transfer

By: Xu, Yan; et al

New Journal of Chemistry (2024), 48(33), 14684-14689.

Scheme 32 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (98)

Suppliers (88)

31-614-CAS-35529986 Steps: **1** Yield: **98%**

1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine; 30 min, 100 °C

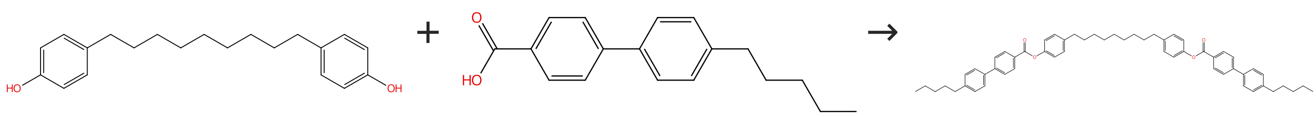
Experimental Protocols

A comprehensive solvent-free approach for the esterification and amidation of carboxylic acids mediated by carbodiimides

By: Traboni, Serena; et al

Tetrahedron (2023), 133, 133291.

Scheme 33 (1 Reaction) Steps: **1** Yield: **98%**



Supplier (1)

Suppliers (64)

31-356-CAS-2895552 Steps: **1** Yield: **98%**

1.1 **Reagents:** 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 4 - 24 h, rt

Experimental Protocols

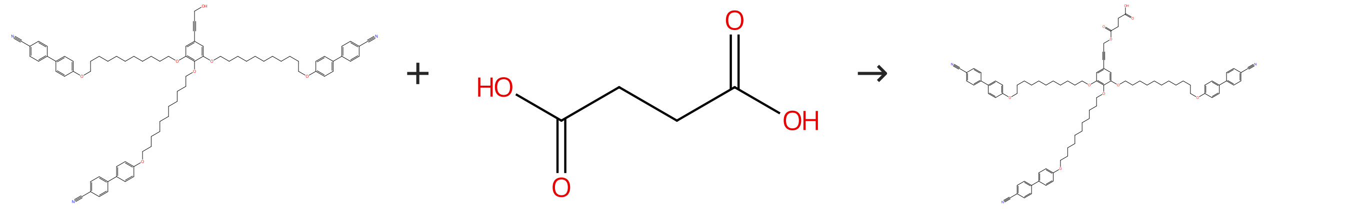
Apolar Bimesogens and the Incidence of the Twist-Bend Nematic Phase

By: Mandle, Richard J.; et al

Chemistry - A European Journal (2015), 21(22), 8158-8167.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **98%**



Suppliers (150)

31-356-CAS-21535741

Steps: **1** Yield: **98%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dimethylformamide, Dichloromethane; 2 d, rt

Experimental Protocols

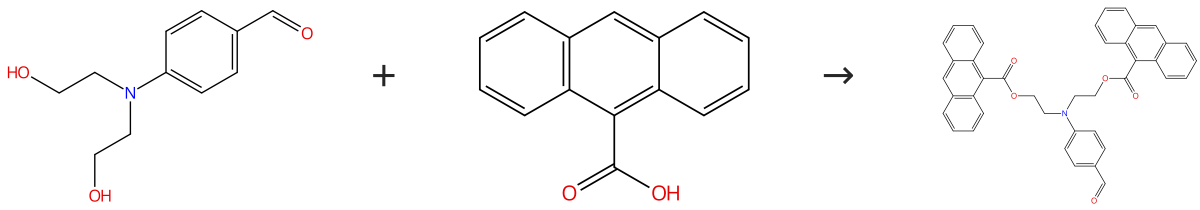
Dendritic Ligands for Magnetic Suspen sions in Liquid Crystals

By: Haehsler, Martin; et al

European Journal of Organic Chemistry (2019), 2019(48), 7820-7830.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **98%**



Suppliers (58)

Suppliers (93)

31-356-CAS-23058794

Steps: **1** Yield: **98%**

1.1 **Reagents:** Triphenylphosphine
Catalysts: Diethyl azodicarboxylate
Solvents: Tetrahydrofuran; 0 °C; 48 h, rt

Experimental Protocols

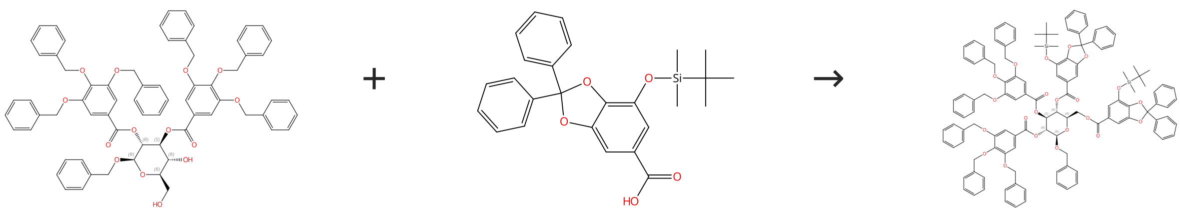
Mild and in situ photo-crosslinking of anthracene-functio nalized poly(aryl ether ketone) for enhancing temporal stability of organic NLO materials

By: Tian, Yanxin; et al

Journal of Materials Science (2021), 56(9), 5910-5923.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **98%**



Absolute stereochemistry shown

Absolute stereochemistry shown

31-356-CAS-11904667

Steps: **1** Yield: **98%**

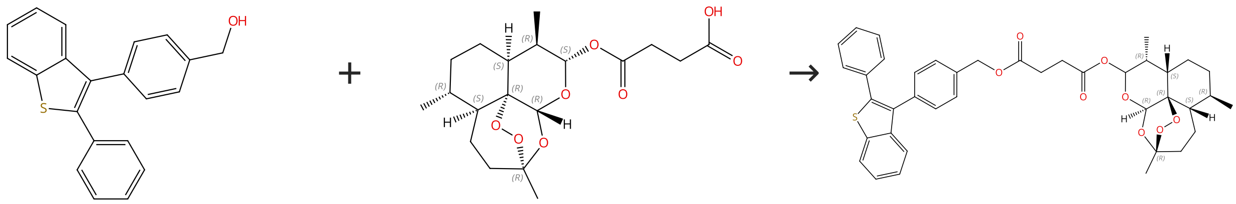
1.1 **Catalysts:** 4-(Dimethylamino)pyridine, 4-Pyridinamine, *N*,*N*-dimethyl-, hydrochloride (1:1)
Solvents: Dichloromethane

Ellagitannin Chemistry. The First Total Chemical Synthesis of an Ellagitannin Natural Product, Tellimagrandin I

By: Feldman, Ken S.; et al

Journal of the American Chemical Society (1994), 116(5), 1742-5.

Scheme 37 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown, Rotation (+)

Suppliers (97)

Absolute stereochemistry shown

31-614-CAS-34328356 Steps: **1** Yield: **98%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; overnight, 0 °C

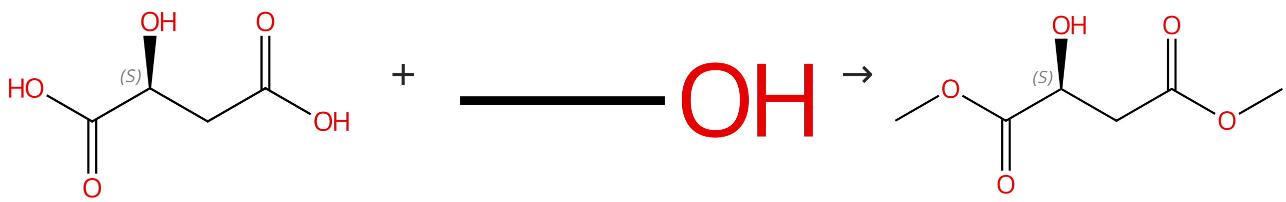
Experimental Protocols

Synthesis of novel artesunate-benzothiophene and artemisinin-benzothiophene derivatives

By: Ozok, Omruye; et al

Natural Product Research (2022), 36(20), 5228-5234.

Scheme 38 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (118)

Suppliers (444)

Absolute stereochemistry shown, Rotation (-)

Suppliers (66)

31-356-CAS-5833476 Steps: **1** Yield: **98%**

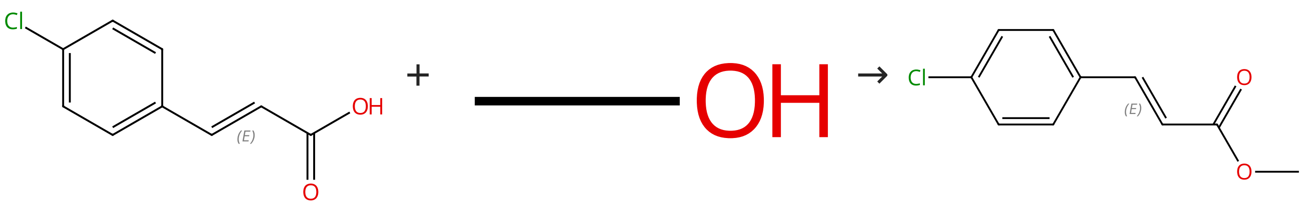
1.1 **Reagents:** Boron trifluoride etherate
Solvents: Methanol; 6 h

First total synthesis of the E- and Z-isomers of cytospolide-D

By: Kamal, Ahmed; et al

Tetrahedron: Asymmetry (2014), 25(2), 148-155.

Scheme 39 (1 Reaction) Steps: **1** Yield: **98%**



Double bond geometry shown

Suppliers (56)

Suppliers (444)

Double bond geometry shown

Suppliers (37)

31-356-CAS-20621274 Steps: **1** Yield: **98%**

1.1 **Catalysts:** Sulfuric acid; 3 - 24 h, reflux

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; neutralized, rt

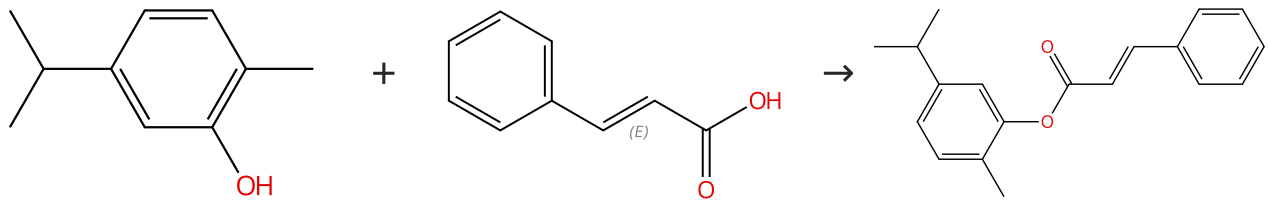
Experimental Protocols

Antimicrobial activity of 4-chlorocinnamic acid derivatives

By: Silva, Rayanne H. N.; et al

BioMed Research International (2019), 3941242pp..

Scheme 40 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (103)

Double bond geometry shown

Suppliers (160)

Suppliers (2)

31-614-CAS-24403401 Steps: **1** Yield: **97%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 24 h, rt

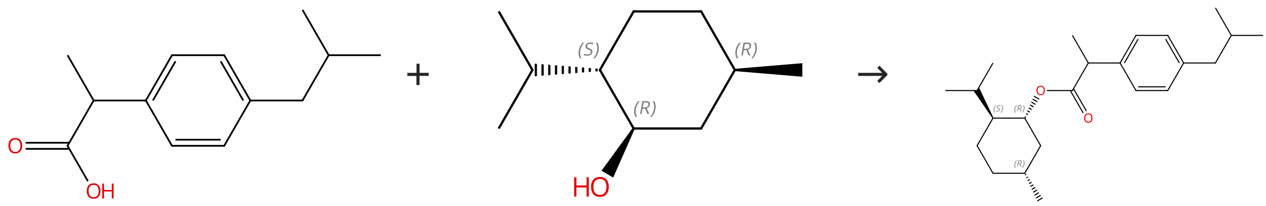
Experimental Protocols

Preparation, spectral characterization and anticancer potential of cinnamic esters

By: Goncalves, Renan O.; et al

Journal of the Brazilian Chemical Society (2021), 32(10), 1931-1942.

Scheme 41 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (146)

Absolute stereochemistry shown, Rotation (-)

Suppliers (132)

Absolute stereochemistry shown

31-356-CAS-3502121 Steps: **1** Yield: **97%**

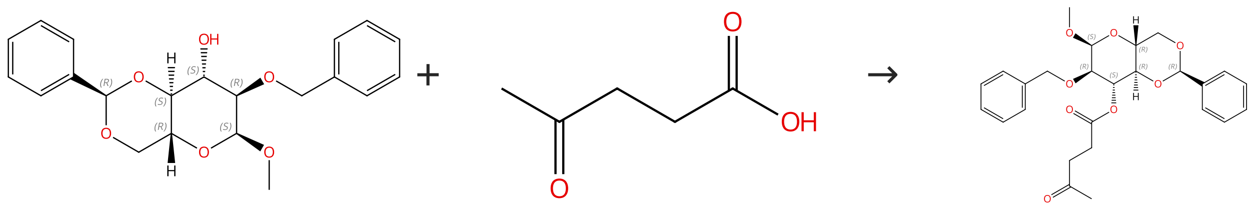
1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Chloroform; 0 °C

Chiral auxiliary-mediated enantioenrichment of (±)-ibuprofen, under Steglich conditions, with secondary alcohols derived from (R)-carvone

By: Amongero, Marcela; et al

Journal of the Brazilian Chemical Society (2010), 21(6), 1017-1036.

Scheme 42 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (+)

Suppliers (5)

Suppliers (98)

Absolute stereochemistry shown

31-614-CAS-35529984 Steps: **1** Yield: **97%**

1.1 **Reagents:** Diisopropylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine; 30 min, 100 °C

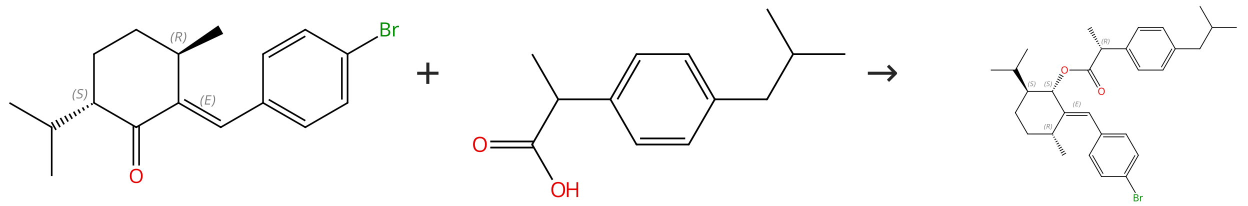
Experimental Protocols

A comprehensive solvent-free approach for the esterification and amidation of carboxylic acids mediated by carbodiimides

By: Traboni, Serena; et al

Tetrahedron (2023), 133, 133291.

Scheme 43 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

Suppliers (146)

Absolute stereochemistry shown, Rotation (-) Double bond geometry shown

31-356-CAS-8037202 Steps: **1** Yield: **97%**

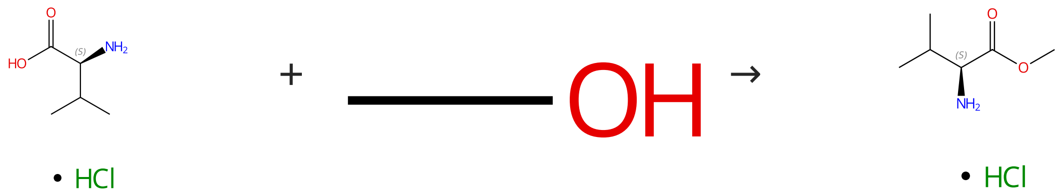
1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Chloroform; overnight, rt

Chiral auxiliary-mediated enantioenrichment of (±)-ibuprofen, under Steglich conditions, with secondary alcohols derived from (R)-carvone

By: Amongero, Marcela; et al

Journal of the Brazilian Chemical Society (2010), 21(6), 1017-1036.

Scheme 44 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (+)

Suppliers (34)

Suppliers (444)

Absolute stereochemistry shown, Rotation (+)

Suppliers (92)

31-356-CAS-21877812 Steps: **1** Yield: **97%**

1.1 **Reagents:** Thionyl chloride
Solvents: Methanol; 0 °C; 4 h, 0 °C

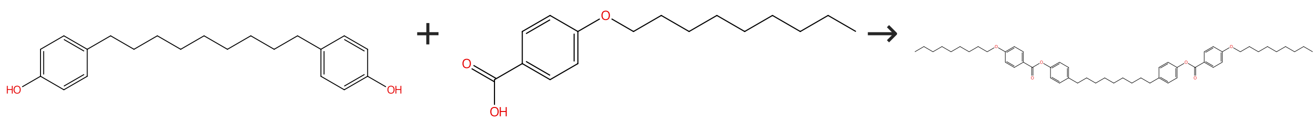
Experimental Protocols

Synthesis and antimalarial activity of (S)-methyl-(7-chloroquinolin-4-ylthio)acetamidoalquilate derivatives

By: Colmenarez, Custodiana; et al

Journal of Chemical Research (2020), 44(3-4), 161-166.

Scheme 45 (1 Reaction) Steps: **1** Yield: **97%**



Supplier (1)

Suppliers (48)

31-356-CAS-14528566 Steps: **1** Yield: **97%**

1.1 **Reagents:** 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 4 - 24 h, rt

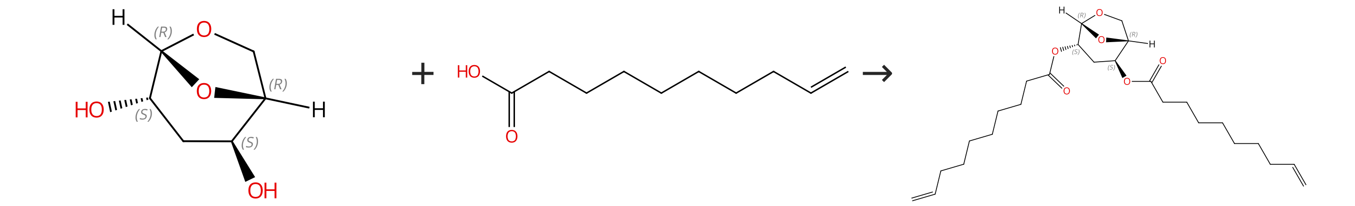
Experimental Protocols

Apolar Bimesogens and the Incidence of the Twist-Bend Nematic Phase

By: Mandle, Richard J.; et al

Chemistry - A European Journal (2015), 21(22), 8158-8167.

Scheme 46 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (3)

Suppliers (64)

Absolute stereochemistry shown, Rotation (-)

31-614-CAS-45857159 Steps: **1** Yield: **97%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; > 1 min, 0 °C; 0 °C → rt; 1 h, rt

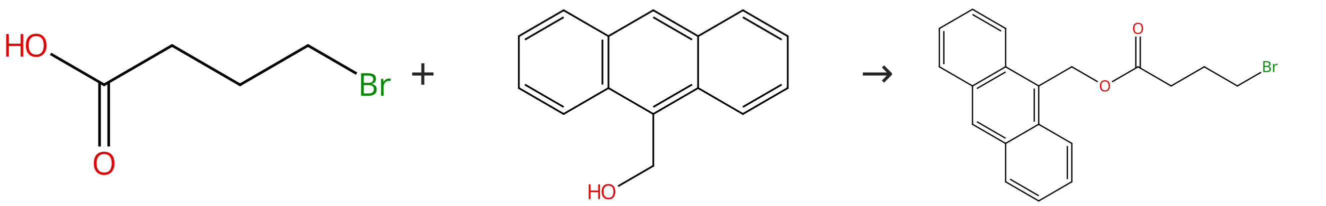
Experimental Protocols

Green Insights into Levoglucosenone Polyesters: ADMET Polymerization vs Metal- and Enzyme-Catalyzed Polycondensation

By: Giri, German; et al

ACS Sustainable Chemistry & Engineering (2025), 13(25), 9553-9564.

Scheme 47 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (83)

Suppliers (83)

31-356-CAS-23059331 Steps: **1** Yield: **97%**

1.1 **Reagents:** Dicyclohexylcarbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane, Tetrahydrofuran; 24 h, rt; 1 h, 0 °C

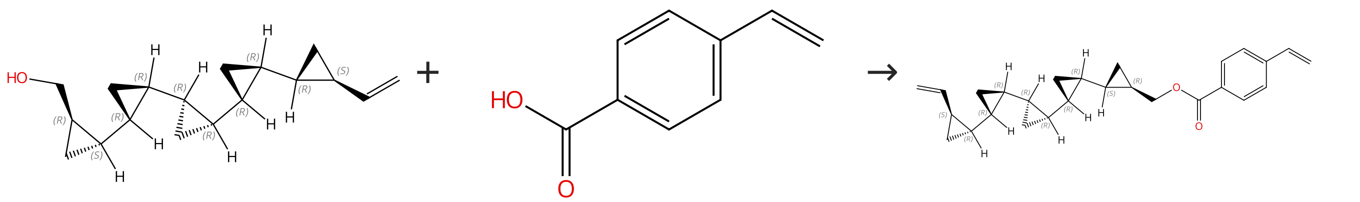
Experimental Protocols

Mild and in situ photo-crosslinking of anthracene-functionalized poly(aryl ether ketone) for enhancing temporal stability of organic NLO materials

By: Tian, Yanxin; et al

Journal of Materials Science (2021), 56(9), 5910-5923.

Scheme 48 (1 Reaction) Steps: **1** Yield: **96%**



Absolute stereochemistry shown

Suppliers (90)

Absolute stereochemistry shown

31-356-CAS-7096530 Steps: **1** Yield: **96%**

1.1 **Reagents:** Dicyclohexylcarbodiimide, 4-(Dimethylamino)pyridine
Solvents: Dichloromethane

1.2 **Solvents:** Water

1.3 **Solvents:** Dichloromethane

Experimental Protocols

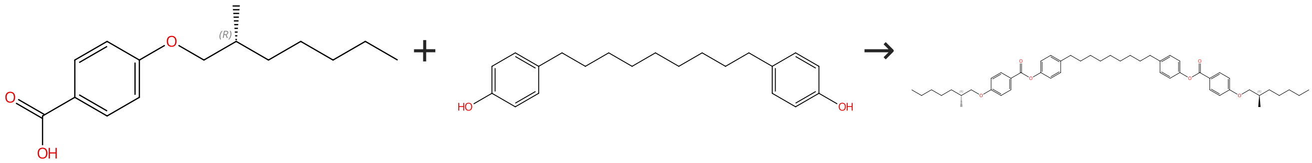
Synthesis and Characterization of Coronanes: Multicyclopropane-Fused Macrocyclic Arrays

By: Barrett, Anthony G. M.; et al

Journal of Organic Chemistry (2001), 66(7), 2187-2196.

Scheme 49 (1 Reaction)

Steps: **1** Yield: **96%**



Absolute stereochemistry shown

Supplier (1)

Absolute stereochemistry shown

31-356-CAS-6162585

Steps: **1** Yield: **96%**

1.1 **Reagents:** 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Dichloromethane; 4 - 24 h, rt

Experimental Protocols

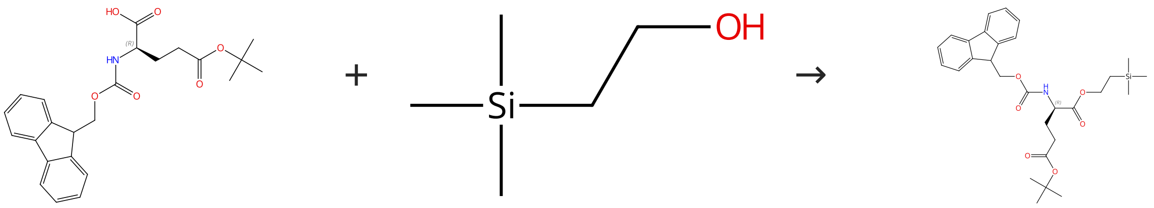
Apolar Bimesogens and the Incidence of the Twist-Bend Nematic Phase

By: Mandle, Richard J.; et al

Chemistry - A European Journal (2015), 21(22), 8158-8167.

Scheme 50 (1 Reaction)

Steps: **1** Yield: **96%**



Absolute stereochemistry shown

Suppliers (80)

Suppliers (85)

Absolute stereochemistry shown, Rotation (+)

31-356-CAS-22375121

Steps: **1** Yield: **96%**

1.1 **Catalysts:** 4-(Dimethylamino)pyridine
Solvents: Ethyl acetate; rt → 0 °C

1.2 **Reagents:** Dicyclohexylcarbodiimide
Solvents: Dichloromethane; 2 h, rt

Modular Total Synthesis of Farnesyl Analogues of Cell Wall Precursors Lipid I and II Containing the Staphylococcus aureus Pentaglycine Bridge Modification

By: Wingen, Lukas M.; et al

Journal of Organic Chemistry (2020), 85(15), 10206-10215.